

# Telecom Customer Churn Analysis

## 1. Project Overview

### Telecom Customer Churn Analytics

This project focuses on analyzing telecom customer data to understand customer churn patterns and identify the key factors influencing customer retention.

The project uses SQL for data preparation, data validation, ETL, and business analysis, followed by Power BI for data modeling, DAX calculations, interactive dashboards, and visualization.

The analysis focuses on customer behavior, contract types, payment methods, tenure, CLTV, churn risk, and churn reasons to generate actionable business insights.

### Project Objective

The main objective is to identify customers and factors associated with high churn, understand customer behavior, and provide data-driven recommendations to help the telecom company improve customer retention and reduce churn.

### Key Areas of Analysis

- Customer churn and churn rate
- Contract and payment method impact
- Customer tenure and churn
- CLTV and churn behavior
- Customer churn risk
- Major reasons for customer churn
- Business recommendations for customer retention

### Tools Used

- **SQL** – Data preparation, validation, ETL and business analysis
- **Power BI** – Data modeling, DAX, visualization and dashboard development
- **Power Query** – Data transformation and preparation

## 2. Dataset & Business Objective

### Dataset

The project uses the **Telco Customer Churn IBM Dataset**, containing customer-level information related to demographics, location, services, contracts, billing, charges, customer lifetime value (CLTV), churn status, and churn reasons.

The dataset contains **7,043 customer records** and was used as the primary source for SQL analysis and Power BI dashboard development.

### Business Objective

The objective of the analysis is to understand **why customers churn, which customer segments have higher churn risk, and what factors influence customer retention.**

The analysis aims to answer key business questions such as:

- Which customers are more likely to churn?
- Which contract and payment methods have higher churn?
- How does customer tenure affect churn?
- Do high-value customers have lower churn?
- What are the major reasons customers leave?
- Which areas should the company focus on for customer retention?

The final goal is to provide **data-driven insights and recommendations** that can help the telecom company reduce customer churn and improve customer retention.

### 3. SQL Staging, Data Cleaning & Validation

#### Staging Table

The raw dataset was first imported into a SQL staging table named `telco_raw`. The staging approach was used to preserve the original data and perform data profiling and validation before loading the data into the final analytical tables.

#### Data Validation

The following checks were performed on the staging table:

- Duplicate customer records
- NULL values
- Blank/empty values
- Inconsistent categorical values
- Data type validation
- Latitude and longitude range validation
- Business logic validation for missing values

#### Key Data Quality Findings

The data validation process showed that most fields contained no NULL, blank, or inconsistent values.

Two important cases were identified:

##### 1. Blank Total Charges — 11 records

All 11 customers had `tenure_months = 0`, indicating that they were newly joined customers who had not generated any charges yet.

Therefore, these values were considered **business-valid** rather than data errors.

During ETL, the blank values were converted to NULL because the target `total_charges` column was defined as `DECIMAL(10,2)`.

##### 2. Blank Churn Reason — 5,174 records

The blank `churn_reason` values belonged to customers whose `churn_label` was No.

Since these customers had not churned, a churn reason was not applicable. Therefore, these blanks were considered **business-valid**.

#### Cleaning Conclusion

The data was validated without removing valid business records. After validation, the cleaned and verified data was prepared for loading into the final dimension and fact tables.

## 4. Database Design & ETL

### Star Schema Design

After data validation, the dataset was organized into a **star schema** to create a structured analytical database.

The final database consists of five tables:

| Table                 | Purpose                                                |
|-----------------------|--------------------------------------------------------|
| dim_customer          | Customer demographic information                       |
| dim_location          | Customer location and geographic information           |
| dim_services          | Services used by customers                             |
| dim_contract          | Contract, billing and payment information              |
| fact_customer_metrics | Tenure, charges, churn, CLTV and churn-related metrics |

The fact\_customer\_metrics table acts as the central fact table, while the dimension tables provide descriptive information for analysis.

### ETL Process

The validated data was loaded from the telco\_raw staging table into the final analytical tables using SQL.

The main ETL steps were:

1. Created dimension and fact tables.
2. Inserted validated customer, location, service and contract data.
3. Loaded customer metrics into the fact table.
4. Converted total\_charges from VARCHAR to DECIMAL(10,2).
5. Converted the 11 blank total\_charges values to NULL during loading.
6. Performed row-count validation after loading.

The final fact table contained **7,043 customer records**, confirming successful data loading.

## 5. SQL Business Analysis

After completing data preparation and ETL, SQL was used to perform business analysis and identify the major factors associated with customer churn. The results were also used to define the KPIs and visualizations for the Power BI dashboard.

### Key Analysis & Findings

| Analysis                | Key Finding                                                              |
|-------------------------|--------------------------------------------------------------------------|
| Total Customers         | 7,043 customers                                                          |
| Churn Customers         | 1,869 customers                                                          |
| Churn Rate              | 26.54%                                                                   |
| Contract Analysis       | Month-to-Month customers showed the highest churn                        |
| Payment Method Analysis | Electronic Check customers had the highest churn rate at 45.29%          |
| Gender Analysis         | Female churn rate: 26.92%; Male: 26.16% — minimal difference             |
| Tenure Analysis         | 0–12 months customers had the highest churn rate at 47.44%               |
| CLTV Analysis           | Low CLTV customers had the highest churn rate at 34.43%                  |
| Churn Reason Analysis   | Customer support and competitor-related factors were major churn drivers |

### Major Business Insights

- Customers with **shorter tenure** are significantly more likely to churn.
- **Month-to-Month contracts** show higher churn compared with longer-term contracts.
- Customers using **Electronic Check** have a higher churn rate than customers using automatic payment methods.
- **Low CLTV customers** have higher churn risk than high CLTV customers.
- **Customer support issues and competitor offers** are among the major reasons for customer churn.
- Gender showed only a minor difference and does not appear to be a significant churn factor.

These SQL findings were used as the foundation for the Power BI dashboard and helped determine the KPIs, customer segments, major churn patterns, and visualizations required for the final analysis.

## 6. Power BI Data Model & DAX

### Data Model

The validated SQL tables were imported into Power BI and a structured data model was created using the star schema developed during the SQL stage.

The model connects the customer dimension with the location, services, contract, and customer metrics data to enable interactive analysis across different customer segments.

### DAX Measures

DAX measures were created to calculate the key performance indicators required for the dashboard.

The main measures include:

- **Total Customers**
- **Churn Customers**
- **Active Customers**
- **Churn Rate**
- **Average Tenure**
- **Average Monthly Charge**
- **Average CLTV**

These measures were used across the dashboard to provide dynamic KPI calculations based on selected filters and slicers.

### Calculated Columns

Additional calculated columns were created to segment customers for deeper analysis:

- **Tenure Group** – Groups customers based on tenure duration.
- **CLTV Group** – Classifies customers into Low, Medium and High CLTV segments.
- **Churn Risk** – Categorizes customers into Low, Medium and High risk based on churn score.
- **Monthly Charges Group** – Groups customers based on their monthly charges.

### Churn Drivers Calculation

To summarize the highest churn rates from different dimensions, additional DAX measures were created for:

- Highest Tenure Churn
- Highest Contract Churn
- Highest Payment Method Churn

A calculated table named **Churn Drivers** was then created using UNION() and ROW() to bring these values into a single table for visualization.

This helped create a consolidated view of the major churn patterns in the Executive Summary page.

## 7. Power BI Dashboard

The final Power BI dashboard was developed to provide an interactive view of customer churn, churn drivers, customer profiles, and risk segments.

The dashboard consists of four pages, with each page focused on a specific business question.

### Page 1 — Churn Insights

#### Purpose:

To provide a quick summary of the major churn patterns and high-risk customer segments.

#### Business Question:

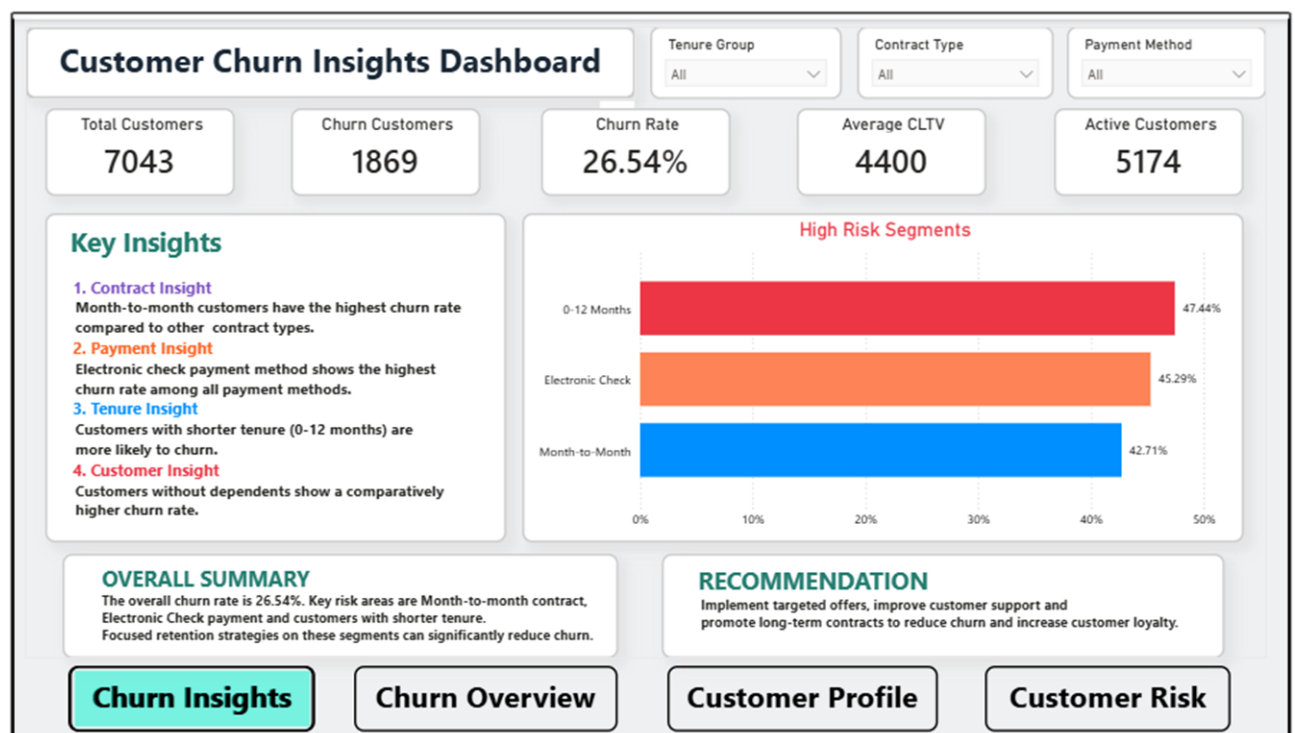
What are the major churn patterns and which customer segments are at higher risk?

#### Answer:

The highest churn is observed among 0–12 months tenure customers (47.44%), Electronic Check users (45.29%), and Month-to-Month contract customers (42.71%).

#### Key Analysis:

- High-Risk Customer Segments
- Major Churn Patterns
- Overall Churn Summary
- Business Recommendations



## Page 2 — Churn Overview

### Purpose:

To provide an overall view of churn across major customer and business attributes.

### Business Question:

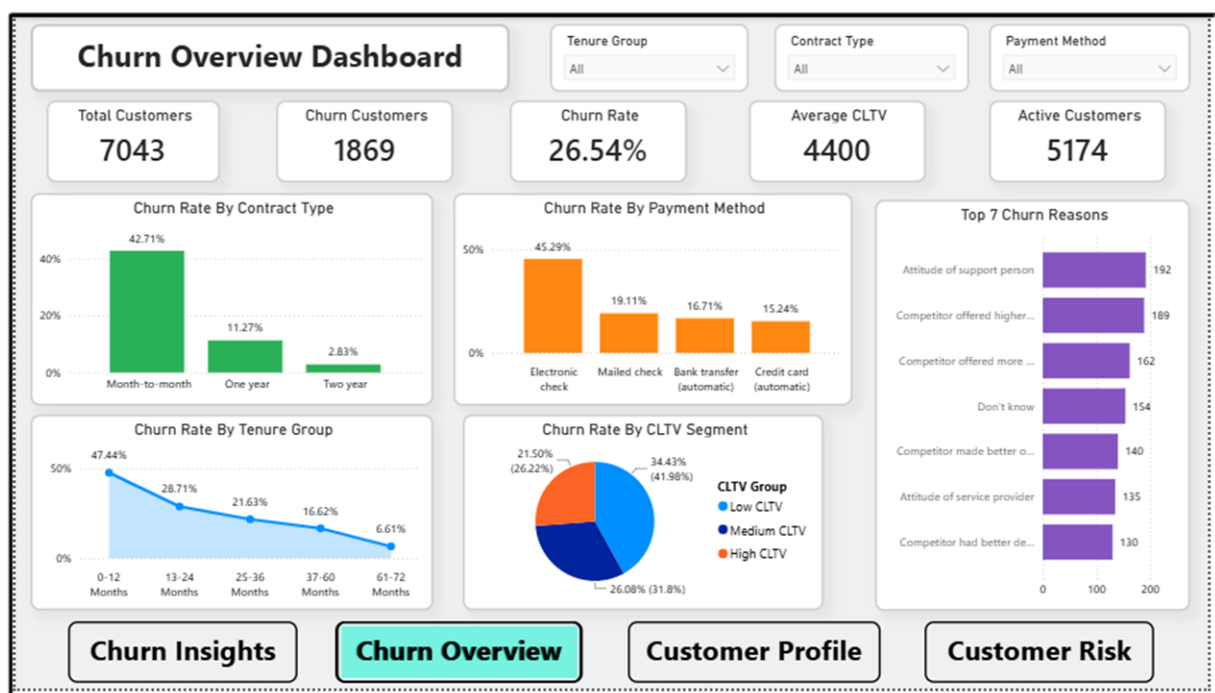
How does customer churn vary across different attributes?

### Answer:

Churn is highest among 0–12 months tenure customers (47.44%), Electronic Check users (45.29%), and Month-to-Month contract customers (42.71%). The leading churn reason is “Attitude of support person” (192 customers).

### Key Analysis:

- Contract-wise Churn
- Payment Method-wise Churn
- Tenure-wise Churn
- CLTV-wise Churn
- Top Churn Reasons





## Page 3 — Customer Profile

### Purpose:

To analyze how customer demographics and subscribed services are associated with churn.

### Business Question:

How do customer profile and services impact churn?

### Answer:

Higher churn is observed among Fiber Optic users (41.89%), Senior Citizens (41.68%), customers without partners (32.96%), and customers without dependents (32.55%). Gender shows only a minor difference.

### Key Analysis:

- Partner-wise Churn
- Senior Citizen-wise Churn
- Internet Service-wise Churn
- Additional Services-wise Churn
- Dependents-wise Churn
- Gender-wise Churn



## Page 4 — Customer Risk

### Purpose:

To identify high-risk customer segments and analyze their churn behavior.

### Business Question:

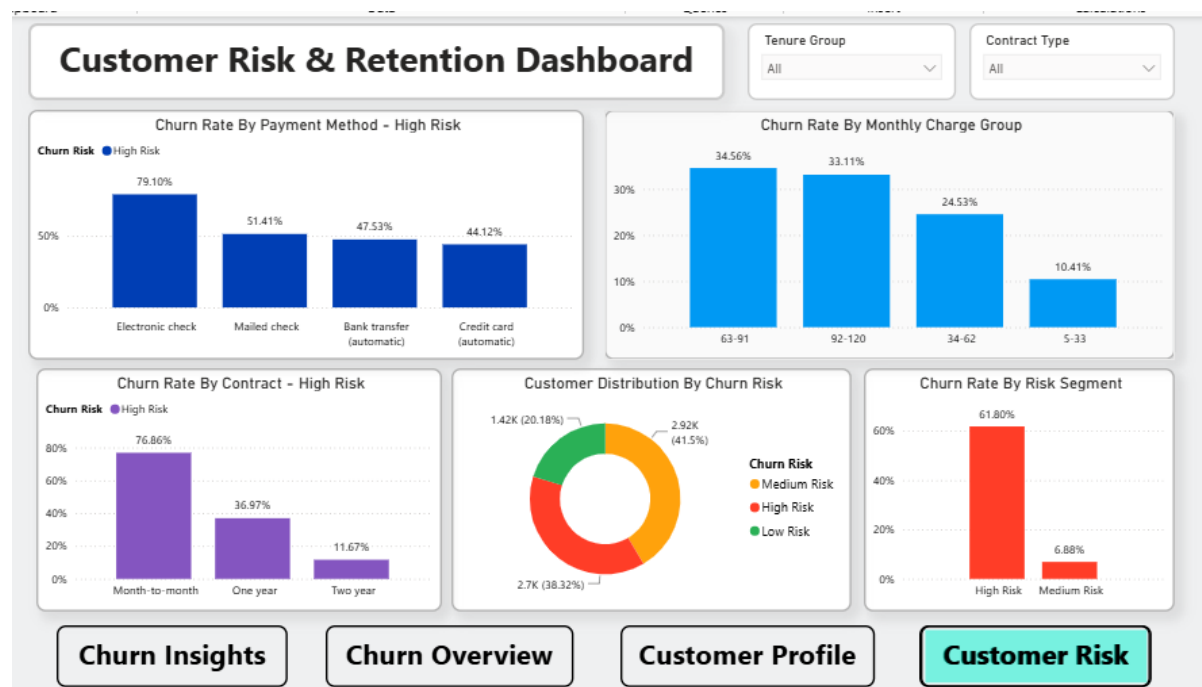
Which customer risk segments have the highest churn?

### Answer:

The High Risk segment has a 61.80% churn rate. Within this segment, Electronic Check users have 79.10% churn and Month-to-Month customers have 76.86% churn.

### Key Analysis:

- High-Risk Payment Method
- Monthly Charge Groups
- High-Risk Contract
- Customer Distribution by Risk
- Churn Rate by Risk Segment



## Interactive Features

The dashboard includes:

- Page navigation buttons
- Bookmarks
- Slicers and filters
- Interactive visuals
- Executive Summary navigation

These features allow users to explore customer churn dynamically from different perspectives.

## 8. Key Business Insights

The analysis identified the following key insights:

- Overall customer churn rate is **26.54%**, with **1,869 customers** having churned.
- Customers with **0–12 months tenure** have the highest churn rate at **47.44%**.
- **Electronic Check** users have a high churn rate of **45.29%**.
- **Month-to-Month** contract customers have a churn rate of **42.71%**.
- The **High Risk** customer segment has a churn rate of **61.80%**.
- **Customer support issues and competitor-related factors** are among the major reasons for customer churn.
- **Fiber Optic** users and **Senior Citizens** also show relatively higher churn rates.

## 9. Business Recommendations

Based on the analysis, the following actions can help improve customer retention:

- **Focus on new customers:** Provide better onboarding, early engagement, and loyalty benefits during the first year.
- **Promote long-term contracts:** Encourage Month-to-Month customers to move to One-Year or Two-Year plans through suitable offers.
- **Improve payment experience:** Encourage Electronic Check users to adopt automatic payment methods.
- **Target high-risk customers:** Use churn risk scores to identify and proactively engage customers who are more likely to churn.
- **Improve customer support:** Address support-related issues and improve response time and service quality.
- **Strengthen competitive offerings:** Provide better pricing, data benefits, speeds, and service packages to compete with alternative providers.

## 10. Conclusion

This project analyzed telecom customer churn using **SQL and Power BI**. The data was cleaned, validated, transformed, and organized into a structured star schema before performing SQL-based business analysis.

Power BI was then used to create the data model, DAX calculations, and interactive dashboards to identify important churn patterns and high-risk customer segments.

The analysis showed that **short tenure, Month-to-Month contracts, Electronic Check payments, and high-risk customer segments** are strongly associated with higher churn. These insights can help the company focus its retention strategies and make more data-driven business decisions.